Haziq M. Khalid

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Education

American University of Sharjah 2023 – 2027
B.S. in Computer Science & B.S. in Mathematics — Double Major Sharjah, UAE

- **Cumulative GPA**: 3.82 / 4.00 | Dean's List & Chancellor's List
- **Coursework**: Advanced Calculus, Machine Learning & Data Mining, Optimization, AI, Linear Algebra, Statistics

Publications

ERGO: Entropy-guided Resetting for Generation Optimization in LLMs
Lead Author | *UncertaiNLP @ EMNLP '25 ; MTI-LLM @ NeurIPS '25*

- Cited in *"Do LLMs Benefit From Their Own Words?"* — Huang et al., MIT & IBM Watson AI Lab, 2026.
- Cited in *"Lost in Stories: Consistency Bugs in Long Story Generation by LLMs"* — Li et al., Microsoft & SUTD, 2026

Noise Steering for Controlled Text Generation: Improving Diversity and...
First Author | *BEA @ ACL 26'*

Experience

Algoverse AI — Presented at UncertaiNLP @ EMNLP in Suzhou, China *Mar 2025 – Aug 2025*
AI Researcher Remote

- Conceptualized and led **ERGO**, a novel model-agnostic inference-time framework applying **entropy-based uncertainty quantification** to detect and resolve LLM context degradation across multi-turn conversations.
- Designed a prompt-reset protocol distilling multi-turn context into optimized single-turn prompts; ran **1,000+ simulated dialogues** achieving a **56.6%** increase in average task accuracy, **35.3%** decrease in performance variation, and **24.7%** increase in best case performance over standard multi-turn baselines.
- Wrote a reproducible evaluation suite across 5 task types spanning open-source (LLaMA, Phi) and proprietary models (GPT), with OOP design, modular pipelines, and CI/CD via GitHub Actions.

American University of Sharjah — Undergraduate Research Grant recipient *Mar 2025 – Present*
Research Assistant Sharjah

- Investigating **creativity induction in LLMs** for EGRA-constrained Arabic children's story generation, a **constrained generation** problem requiring models to produce imaginative output within linguistic and pedagogical boundaries.
- Conducting parallel research on Human-Computer Interaction (HCI) and long-horizon LLM conversation reliability, extending ERGO's **uncertainty-aware** framework to real-world deployment settings.
- Awarded a **competitive Undergraduate Research Grant** for Fall '25, one of few solo projects selected campus-wide.

Projects

Intel Technical Assessment — MiniCPM-o Model Compression *Dec 2025*
Python · PyTorch · Model Pruning · Low-rank Decomposition · Architecture Engineering · GitHub Actions

- Achieved a **97.6% model size reduction** (250 MB → 6 MB) via structured pruning and low-rank decomposition; reconstructed tokenizer assets from scratch to implement a **1,024-token vocabulary** (92% footprint reduction).
- Modified source architecture to bypass hard model constraints; approach independently **verified by Intel's technical lead** via automated GitHub Actions forward-pass testing.

Super-Weight Aware INT4 Quantization Framework *Nov 2025*
Python · PyTorch · HuggingFace · Model Compression · Transformers

- Reproduced and extended Apple's super-weight quantization technique, identifying **1–6 critical scalar weights** per LLM layer that disproportionately impact output quality across transformer architectures (OLMo family).
- Implemented INT4 quantization with super-weight preservation via activation spike detection; achieved **competitive perplexity** with SOTA methods at larger block sizes — **no calibration dataset required**.

Skills & Technologies

Languages: Python, C++, JavaScript, SQL, HTML/CSS, Node.js

Frameworks & Libraries: PyTorch, FastAPI, Playwright, HuggingFace, OpenCV, BeautifulSoup, PRAW

Infrastructure & DevOps: Git, GitHub Actions (CI/CD), Docker, AWS, Google Cloud, REST APIs, OOP Design, TDD, Microservices

AI/ML: Transformers, LLMs, Uncertainty Quantification, Inference-time Optimization, Constrained Generation, INT4 Quantization, Model Pruning, Computer Vision

Academic & Research: Published AI Researcher (NeurIPS '25, EMNLP '25) ; Technical Team Member, Google Developer Groups on Campus